

Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

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Abstract

A language model is trained to predict text and should be judged by how well it predicts the text it will be used on. Base models are instead chosen by public benchmarks, which vendors optimise for, which can leak into training data, and which score a model before the adaptation every team applies. We return to the model’s own objective: adapt every candidate to fresh, decontaminated domain text by the same procedure and rank by held-out bits per byte—no labels, minutes of GPU time per model with a fixed recipe. Adaptation changes the ordering. With no ground truth to appeal to, we compare the new ordering with independent views of the models—fine-tuning fifteen of them on a news task, and quizzes built from four corpora. Each benchmark falls short on some view; adapted bits per byte agrees with every one at least as closely as any benchmark, matching the fine-tuned outcome in 0.949 of size-matched pairs. It also tracks HellaSwag and turns toward the reasoning benchmarks on mathematics. Code and artifacts are released.

1 Introduction

Everything a base model can do, it learned by predicting the next piece of text. The most direct way to ask how good one is, for a given use, is therefore to measure how well it predicts the text of that use. That is not how base models are chosen. A new one ships most weeks, and what arrives with it is a scorecard—MMLU-Pro, HellaSwag, GSM8K—read as the answer to *how good is this model, for what we do?* This paper argues that language models should be evaluated as language models: by the loss they were trained to minimise, on the text they will be used on. No task-specific score can serve as ground truth for a base model, which is chosen before most of its uses exist; the case for loss is that it is the one quantity every use is built on, and the evidence is that it agrees with every concrete use we could test at least as closely as any benchmark, while each benchmark falls short on some.

The scorecard misleads for two reasons. The first is that benchmarks are public. Each was written by other people, for other purposes, and released before the model existed; vendors optimise for the ones they report, and their items and subject matter can reach pre-training data. A high score is therefore consistent with a genuinely stronger model or with one that saw more of that exam, and because pre-training mixtures are not disclosed, no reader can tell which. Inside our own cohort, Llama-3.2-1B and Qwen2.5-1.5B are 2.1 points apart on HellaSwag and 3.4% apart in adapted bits per byte, yet 55.3 points apart on GSM8K: two models that model fresh English almost identically well differ by nearly a factor of ten on one exam. The second reason is timing. Teams rarely deploy a base model exactly as it ships; they adapt it to their data first, so what matters is how good a model *becomes*. A model that has simply never seen prose like yours can look weak out of the box and still overtake one that looked stronger. Gemma-4-12B is the case in point: of fifteen models it is ninth on GSM8K, eighth on MMLU-Pro and thirteenth on raw predictive loss, and it builds the best fine-tuned system of all (Table 2).

There is no universal ground truth for which base model is “better”; any yardstick is a choice, and loss is the principled one. Every base model was trained to minimise cross-entropy on text, so held-out loss is

each model’s own objective measured out of sample, and every other ability it has was learned through that objective. Cross-entropy is also a proper scoring rule: in expectation over text drawn from the domain’s distribution p , a model q scores $H(p, q) = H(p) + \text{KL}(p \| q)$, and $H(p)$ is the same for every model, so the difference between two models’ expected losses is *exactly* the difference in how far each is from the distribution that produced the text; held-out loss estimates it [1]. Taking that seriously asks three things of the measurement, and each answers one of the failures above. The loss is counted in *bytes*, because every model counts in its own tokens and bytes are the one unit they all share. It is computed on *fresh, decontaminated* text—dated after every candidate shipped, deduplicated, and screened for membership—so that no candidate has trained on it. And it follows *equal adaptation*: every candidate is adapted to the domain by the same procedure—its own hyperparameter search, then training to convergence—which turns “which model already writes like my corpus” into “which model becomes best at it” and so measures each model after adaptation, as it will be used (Figure 1). The protocol needs only the domain’s raw text, no labels, and minutes of GPU time per model. Train-before-test [15] also prepares candidates identically before comparing them, but with labelled benchmark data; we do it unsupervised, on fresh text, and in bytes.

We test the protocol on seventeen base models from eight families (0.5B–35B) and four corpora, and the results form one argument. **Adaptation changes the ordering.** Against a fill-in-the-blank quiz on held-out news, eleven of fifty-five model pairs move from disagreeing to agreeing with it after adaptation and none the other way, because the unadapted number mixes familiarity with this prose into capability and adaptation largely removes the familiarity (Section 3.1). **The new ordering recovers what benchmarks measure, and can be aimed.** Over eleven models it matches HellaSwag at Spearman 0.96–0.98 on news, Reddit and Hacker News, up from 0.63–0.72 before adaptation, and when the text is mathematics it swings to GSM8K and MMLU-Pro (0.91 and 0.93; Section 3.2). **It anticipates a real use it never saw.** Measured before any task exists, it matches the ordering of fifteen models fine-tuned on a news task in 0.949 of size-matched pairs (within $2\times$ in parameters)—more than any benchmark (HellaSwag 0.872; GSM8K and MMLU-Pro near a coin)—and places every model within one rank of where it finishes (Section 3.3). **And it holds.** Over seventeen models and four corpora, against a quiz built from each corpus, it never agrees less than a public benchmark does (Section 3.4). Section 2 gives the protocol and the independent views we compare it with, Section 3 the results in this order, and Section 4 how to use them. Every number recomputes from a committed artifact: <https://github.com/DavidAfonsoValente/entropy-bench>.

2 The Protocol

The protocol has two halves, shown in Figure 1: the measurement itself, and the independent views of the models we compare it with.

2.1 The measurement

The measurement runs in three phases, and the order is load-bearing: the corpus is fixed and screened before any model touches it.

Phase 1: a decontaminated corpus. Evaluation text must be text no candidate has seen, and we guarantee it three ways. *Time*: collect text from the target domain dated after every candidate’s release—Reddit and Hacker News are scraped in August 2026, after every model in the cohort shipped, and the arXiv set runs from 9 June to 20 August. *Hygiene*: deduplicate exactly, filter with Gopher quality heuristics, remove MinHash/LSH near-duplicates at Jaccard 0.60, and split 80/10/10 with no identifier or normalised-text overlap across splits, so nothing in the test split reappears in training. *Membership screens*: for the news snapshot (Google News, 8 June 2026, 119,054 articles), which predates part of the release window, every

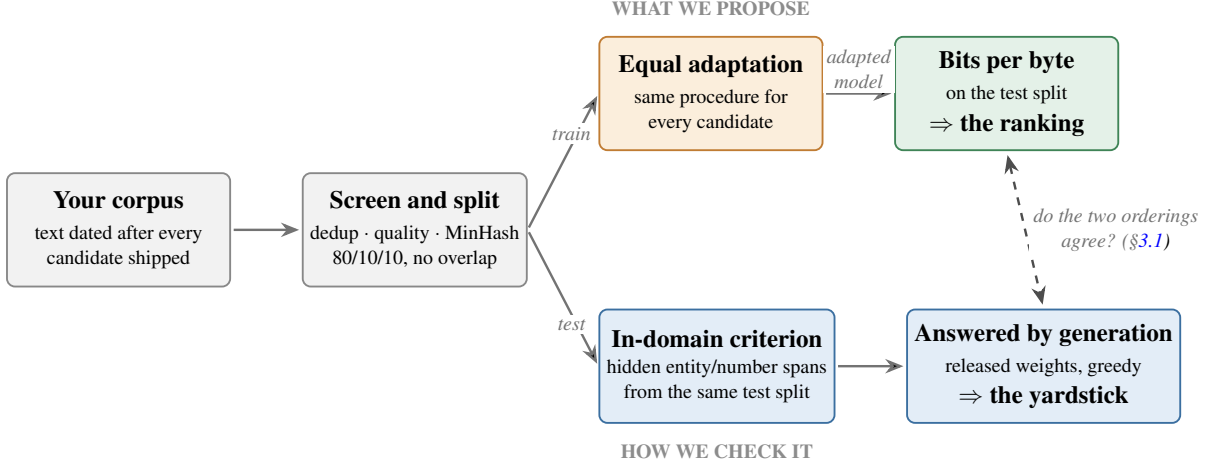


Figure 1: The whole study on one page. The upper arm is the protocol we propose: take text from your own domain that postdates every candidate, screen and split it, adapt every candidate by the same procedure, and rank by bits per byte on the held-out test split. The lower arm is how we check that ranking without appealing to the benchmarks under criticism: the *same* held-out text supplies a criterion whose items are answered by generation from the released base weights, so it is neither trainable-for nor scored by the quantity it validates. The paper is the comparison of the two orderings.

model is run through three reference-free membership tests—Min-K++ [10], sensitivity to in-corpus context, and a perturbation test—and the *union* of everything any model flags is quarantined, so every candidate is measured on one identical retained corpus (Appendix B).

Phase 2: equal adaptation. Every candidate is adapted to the retained training text, packed into contiguous 512-token blocks, with LoRA [25] on all linear layers, and every candidate is adapted by the same procedure. Its hyperparameters come from its own Optuna search [24]: a TPE sampler proposes the learning rate, batch size, warm-up, weight decay, LoRA rank, α and dropout; successive halving prunes weak trials from 50 steps on; and every trial is scored by BPB on the validation split, never the test split. Every model receives the same number of trials: a search capped by wall-clock time instead lets small models finish more trials than large ones and so tunes them better, a size bias we observed directly (Appendix D). The chosen configuration is then trained until validation BPB stops improving, so each candidate is measured near the best it can reach on the domain. The released pipeline implements this procedure, and accepts a hand-picked configuration in place of the search.

Searching seventeen models of up to 35B parameters in full was beyond our compute budget, so the results reported here use that second path: one hand-picked configuration for every candidate—LoRA rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32—and a fixed 250 update steps rather than training to convergence. That is 4.1M training tokens per candidate, which on one A100 cost 3.7 GPU-minutes at 0.5B, 11.6 at 7.7B, 24.7 at 12B and 62.7 at 31B. The configuration sits where our own earlier search settled—rank 16 won for eight of the eleven models and the median winning learning rate was 1.1×10^{-4} (Appendix D)—but it suits no architecture perfectly, and the full procedure can only tune each candidate closer to its best.

Phase 3: bits per byte. Compute cross-entropy on the held-out test split and divide by the UTF-8 byte length $|B|$ of the text those predictions cover:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln 2} \times \frac{|\text{Tokens}|}{|B|} \quad (1)$$

The byte count is a property of the text and identical for every candidate, so a larger vocabulary earns no advantage the way it would under per-token perplexity. Test blocks are independent and non-overlapping, so context resets at every cut, and injected document-start markers stay available as context but are always masked as prediction targets. We report both endpoints, before adaptation and after, because the difference between them is itself informative (Section 3.1).

Cohort and setup. We evaluate 11 base checkpoints spanning dense Transformers, one sparse MoE and one Liquid architecture (0.5B–35B parameters; Table 12), extended to 17 in Section 3.4, from the Llama, Qwen, Gemma, Ministral, LiquidAI, Falcon, Mistral and SmoLLM families [26, 27, 28, 29, 30]. For comparison every model is also scored on MMLU-Pro [18], HellaSwag [19] and GSM8K [20] with *lm-evaluation-harness* 0.4.12 [12] on one host and one build, under the protocol of Appendix G—directly observed scores rather than vendor-reported card values, with the harness logs published so each recomputes from raw output. Adaptation ran on Leonardo (CINECA), 3 nodes $\times$ 4 A100 64 GB. Intervals never resample pairs, because pairs that share a model are not independent: a *selector*’s accuracy resamples **models** (20,000 draws, percentile, clustered), the *criterion*’s own scores its 2,500 **items**, paired across models, and a fine-tuned pair its 500 evaluation articles.

2.2 Independent views, not ground truth

No measurement of a base model can be ground truth, because a base model is chosen before most of its uses exist and each use can order the models differently. A task shows which model is better *at that task*; GSM8K and HellaSwag disagree precisely because they are different tasks. The case for adapted BPB therefore rests on the principle of Section 1, and what can be tested is whether a quantity chosen on principle agrees with concrete uses it was never shown. We build two such views, neither of which uses a public benchmark, and throughout the paper a selector is “right” on a pair when it orders the pair as the view does.

The first is one real use: fine-tune every model on a news task and score the systems that come out (Section 3.3). If that task were the only one that mattered, fine-tuning on it would be the better evaluation; it is expensive, needs labels, and speaks for one task, so we run it once, on fifteen models, as a use the measurement is asked to anticipate. The second is cheap and repeatable: a fill-in-the-blank quiz built from each corpus’s held-out test split: hide a name or a number in an article no model was adapted on, show the model the 1,200 characters before it, and count how often it writes the missing words (Appendix B shows an item). A hidden span that already appears earlier in the prompt is excluded, so a right answer means knowing, not copying. We build one quiz per corpus and call the ranking it produces *the criterion*. A base model has no convention for stopping, so after the answer it often keeps going, or opens with a word of its own; *strict* scoring therefore counts an answer only if the generation begins with it, and *lenient* scoring if it begins within the first 24 characters. Lenient is our primary convention, because it scores what the model knows rather than how it stops. We report both, and every comparison with a public benchmark holds under both: adapted BPB never ranks below one under either. On news, the corpus where we ran both checks, the quiz and the fine-tune rank the fifteen models at Spearman 0.921 (lenient), so on news the two views largely agree, and the quiz carries the comparison to the other three corpora.

The quiz is a fair judge for three reasons. It cannot have been prepared for: Reddit and Hacker News were scraped after every model’s weights were frozen, most of the arXiv set was written after them too, and the news snapshot is protected by the membership screens of Phase 1 (Table 4). It is scored by generation and string match rather than likelihood, which puts distance between it and BPB; HellaSwag, by contrast, ranks continuations by length-normalised conditional log-likelihood—the same object BPB measures—so agreement between BPB and any likelihood-scored criterion would be convergent validity rather than evidence, though greedy decoding is still the argmax of the same conditional. And, most importantly, it is

answered by the released weights, not the adapted ones: whatever the criterion and BPB share, they cannot share the adaptation under test.

3 Results

The results follow the four steps of the argument. The **original eleven models** have both BPB readings on all four corpora and carry Sections 3.1 and 3.2 and the four-corpus comparison of Section 3.4 (55 pairs, 10–11 within $2\times$ in size); the fine-tune uses the fifteen of the **extended seventeen** that fit one GPU, and the size-matched comparison of Section 3.4 all seventeen. Each table names its population.

3.1 Adaptation changes the ordering

Adapting every model before measuring does not just lower every score; it reorders the models, toward the criterion. Table 1 puts each model’s rank under the criterion beside its rank under the two BPB readings. Ranking by the adapted number reproduces the criterion’s ordering almost exactly: seven of eleven models land in precisely the right position and the mean rank error is 0.36 places. The unadapted reading places two of eleven correctly and is off by 2.00 places. Eleven of the fifty-five model pairs flip from wrongly to correctly ordered when the reading changes, and none flips the other way. That gap is not an artefact of a noisy yardstick: resampling the criterion’s own items while holding both BPB orderings fixed leaves the adapted reading at 0.36 [0.00, 0.55] places against 2.00 [2.00, 2.18] for the unadapted one, and the difference between them, 1.64 [1.45, 2.00], excludes zero. These are lenient ranks; under strict scoring the two readings tie in the size-matched band (0.850 each on the seventeen-model cohort, 0.727 on the eleven), so the case for adapting rests on the mechanism as much as on the count.

The mechanism is familiarity. Zero-shot loss conflates two things a practitioner needs separated: *this model already writes like my corpus* and *this model is a capable model*. Figure 2 shows the confound in the raw loss, resolved by position inside a packed evaluation block. Because blocks are independent, every block is a cold start. Unadapted, Gemma-4-12B opens a block at 2.83 bits per byte where Qwen-2.5-1.5B—eight times smaller—opens at 2.04, and it stays behind for roughly the first half of the block; by the end of the block it has passed it, 0.72 against 0.77. A block-averaged number therefore reports mostly how quickly a model finds its footing in unfamiliar prose, which is a real property but not the one the decision turns on. A short equal adaptation supplies exactly the missing familiarity, and what is left is closer to capability. If that is right, the models the unadapted reading most under-rates should be the ones adaptation helps most, and they are: the three largest zero-shot-to-adapted reductions in the cohort are Gemma-4-12B, Gemma-4-31B and LFM2.5-1.2B, which are also the three the free reading most under-rates. Across the cohort the correlation between a model’s adaptation gain and its zero-shot rank error is 0.672; against its *adapted* rank error it is 0.024. The confound shows up in the unadapted number and not in the adapted one.

The cold start is part of the story, not all of it. Scoring every model zero-shot while discarding the first K positions of each block lifts in-band accuracy from 0.545 to at best 0.636 at $K=128$, a quarter of the way to 0.909, the adapted reading’s score on the same eleven models, so the criterion and adapted BPB do not merely meet in a warmed-up regime (Appendix D). Two models show the effect at its sharpest. **Gemma-4-12B** is ninth of eleven on zero-shot BPB but second on both the criterion and adapted BPB—a 12B model the free reading puts below two models under 2B. **LFM2.5-1.2B** is the same effect at the other end: dead last zero-shot at 1.34 bits per byte against a 0.5B Qwen’s 0.93, yet ahead of it after adaptation and on the criterion—a non-Transformer penalised hardest by a metric that rewards familiarity.

Model	Independent criterion			Rank by	
	score	95% CI	rank	zero-shot BPB	adapted BPB
Gemma-4-31B	0.213	[0.197, 0.229]	1	4	1
Gemma-4-12B	0.186	[0.171, 0.201]	2	9	2
Qwen-3.5-35B-MoE	0.147	[0.134, 0.161]	3	1	3
Ministral-3-14B	0.142	[0.128, 0.155]	4	2	4
Qwen-2.5-7B	0.121	[0.108, 0.134]	5	5	6
Qwen-3.5-9B	0.116	[0.104, 0.129]	6	3	5
Llama-3.2-1B	0.104	[0.092, 0.116]	7	7	8
Qwen-3.5-4B	0.101	[0.089, 0.112]	8	6	7
Qwen-2.5-1.5B	0.089	[0.078, 0.100]	9	8	9
LiquidAI-LFM2.5	0.078	[0.067, 0.088]	10	11	10
Qwen-2.5-0.5B	0.066	[0.057, 0.076]	11	10	11
<i>mean rank error</i>				2.00	0.36
<i>ranks exactly right</i>				2 of 11	7 of 11

Table 1: What the adaptation step buys. The eleven-model cohort on the *news* corpus, ordered by an in-domain criterion built from that corpus’s own held-out text and answered by the released base weights. Ranking by adapted BPB reproduces that ordering almost exactly; ranking by the unadapted reading does not. Bold marks an exact match with the criterion. Gemma-4-12B is the clean case: ninth by the free number, second by the criterion, second by the adapted number. Scores are lenient. Intervals are a paired bootstrap over the 2,500 shared cloze items; three of the ten adjacent pairs—Qwen-3.5-35B-MoE over Ministral-3-14B, Qwen-2.5-7B over Qwen-3.5-9B, and Llama-3.2-1B over Qwen-3.5-4B—are not separated by the criterion at all, and nothing in this paper turns on their order. For scale, these are mixed entity-and-number items: within the news corpus numeric spans are answered at 0.279 and entity spans at 0.180 (Appendix B), so 0.213 is a strong score on this set rather than a weak one on an easy one.

3.2 The new ordering recovers what benchmarks measure, and can be aimed

A new measurement should first recover what the existing ones get right. Of the three public benchmarks, HellaSwag is the one closest to language modelling: it asks which ending of a passage is most natural, and the harness scores it by the likelihood of that ending. Before adaptation, BPB on news, Reddit and Hacker News ranks the eleven models at Spearman 0.63–0.72 with HellaSwag; after adaptation the same rank correlation is 0.96–0.98, and on news the adapted ranking orders 53 of HellaSwag’s 55 pairs the same way (Table 6; benchmark scores in Table 8). The effect is specific to what the text is about. On general text the correlation with GSM8K falls on all three corpora, and with MMLU-Pro on two of them; adapted on arXiv mathematics instead, BPB swings toward the reasoning benchmarks—Spearman 0.91 with GSM8K and 0.93 with MMLU-Pro, up from 0.76 and 0.74—so the adaptation corpus steers which capability the ranking tracks. Because HellaSwag and BPB are both likelihoods, this is consistency rather than validation: measuring on fresh text of one’s own gives up nothing the likelihood benchmark captures, and the choice of text decides which capability the ranking follows. Whether it also anticipates a real use of the models is the next question.

3.3 It anticipates a real use it never saw

A measurement made with no task and no labels is most useful if it anticipates how models fare when a task does arrive. So we gave the models one, trained them on it, and asked how often each way of choosing, fixed beforehand, orders the resulting systems as the task does.

Unlike the quiz, this outcome does not live in the text BPB was measured on. We fine-tuned fifteen of the seventeen models—every one that fits a single GPU—on a real in-domain task, writing the opening paragraph of a news article from its body, with one identical LoRA recipe and ROUGE-L against the human-

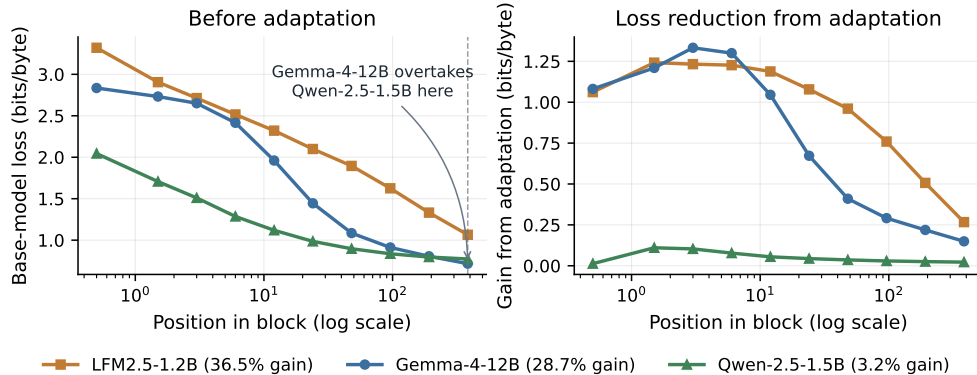


Figure 2: Why the unadapted number mis-ranks: it is largely scoring cold starts. Base loss by position within a packed 512-token evaluation block. Unadapted, Gemma-4-12B opens a block at 2.83 bits per byte against Qwen-2.5-1.5B’s 2.04—a 12B model losing to a 1.5B one—and closes it at 0.72 against 0.77. A block average over-weights the opening, and that is *part* of what puts Gemma-4-12B ninth of eleven on zero-shot BPB (Table 1): discarding the opening outright recovers about a quarter of the gap between the two readings and no more (Appendix D). *Right*: the loss reduction adaptation buys at each position is largest in the opening but does not vanish after it—adaptation teaches the corpus’s opening conventions, and does more besides.

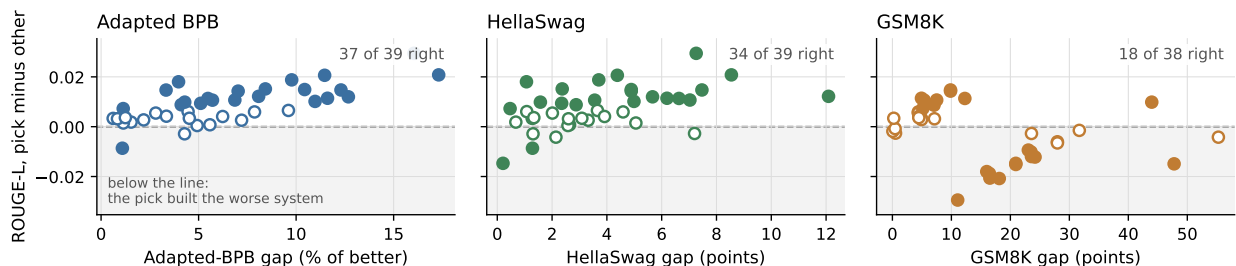


Figure 3: Does each selector agree with the fine-tuned outcome? Every size-matched pair of the fifteen fine-tuned models, oriented so the selector prefers the first: x is how strongly it prefers it, y how much better that model’s fine-tuned system scored. Above the line is a pair the selector got right; filled markers are pairs the fine-tune itself resolves. Adapted BPB is right on 37 of the 39 pairs, and on 21 of the 22 the fine-tune separates; HellaSwag follows at 34. GSM8K is right on 18 of 38, and every one of its eleven gaps between 15 and 25 points sits *below* the line, on a pair where it picked the worse system.

written lede on 500 held-out articles from a split neither the criterion nor the BPB test blocks touch. The analysis was fixed in code before the jobs ran (Appendix E).

Over all 39 size-matched pairs the fifteen systems form, the adapted ranking agrees with the fine-tuned outcome more often than any other way of choosing. Adapted BPB orders 0.949 of the pairs correctly, $[0.817, 1.000]$, ahead of HellaSwag at 0.872, while GSM8K and MMLU-Pro sit at 0.474 and 0.513—a coin (Figure 3, Table 10). Its lead over parameter count is statistically resolved under this recipe, $+0.231$ $[+0.059, +0.452]$, and on the pairs the fine-tune itself separates it picks the better system in 21 of the 22. A deliberately different recipe—four times the rank, three times the learning rate, half the batch, a new seed—leaves the systems’ ordering almost unchanged (Spearman 0.982) and adapted BPB still first at 0.923, right on all 20 pairs that recipe separates. Table 2 shows what this means for a single model on this task. A model’s benchmark standing is a poor guide to where it finishes, while adapted BPB places all fifteen within one rank of their final order. **Gemma-4-12B**, 9th and 8th of fifteen on GSM8K and MMLU-Pro, builds the best system of the fifteen, beating Ministral-3-14B by $+0.018$ $[+0.011, +0.025]$; and the 55.3-point GSM8K gap between Llama-3.2-1B and Qwen-2.5-1.5B buys no measurable difference once both are fine-tuned. Every model improved on its own base weights and cleared what copying the article’s first sentence

Model	Out of the box			After	Adapted BPB
	raw loss	GSM8K	MMLU-Pro	fine-tuning	predicted
Gemma-4-12B	13	9	8	1	2
Llama-3.2-1B	10	15	15	11	11
LFM2.5-1.2B	15	12	13	14	14

Table 2: How a model looks is not how it finishes. Rank among the fifteen fine-tuned models (1 is best). The benchmarks misjudge where these models end up, and raw loss misjudges Gemma-4-12B by twelve places: it is mid-table on both benchmarks and near the bottom on raw loss, then builds the best system of all. Llama-3.2-1B is last on both benchmarks and finishes eleventh. Adapted BPB, measured before any task fine-tuning, places every model within a single rank of where it finishes—all fifteen, and eleven exactly.

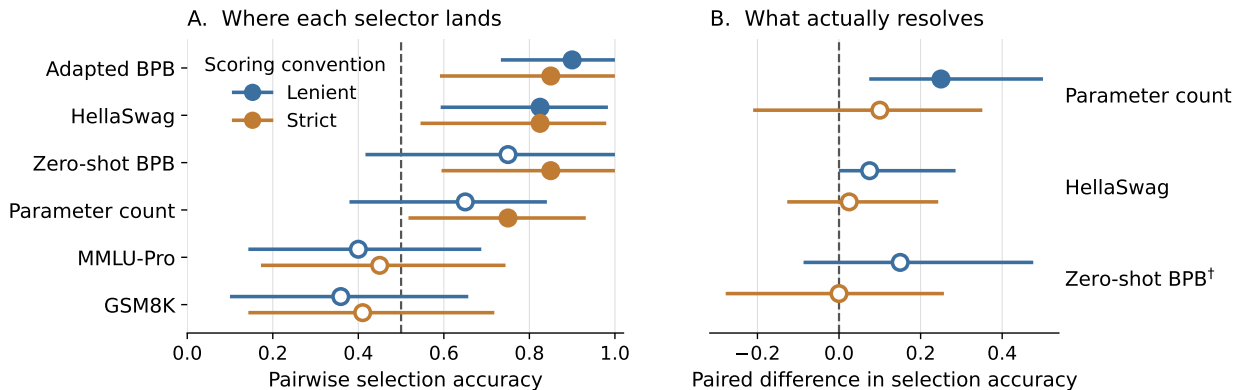


Figure 4: How often each way of choosing a base model agrees with the news quiz. Seventeen models, the 40 pairs within $2\times$ of each other in size, judged by the news quiz (Section 2.2; lenient unless marked strict). **A:** each selector’s accuracy. Adapted BPB is top under lenient scoring and level with the unadapted reading under strict; both generative benchmarks fall below chance by point estimate. **B:** adapted BPB’s accuracy *minus* each alternative’s on the same pairs, so positive favours adapted BPB. Intervals are 95% cluster-bootstrap over models; a **hollow marker** means the interval still contains chance (A) or zero (B). One comparison resolves at this cohort size: adapted BPB over parameter count, $+0.250$ [$+0.074$, $+0.500$], lenient. † computed after the fact rather than preregistered.

scores (0.125).

3.4 It holds on the hardest pairs and on every corpus

The hardest choice is between candidates of similar size, where size stops deciding. Eleven models give only eleven pairs within $2\times$ of each other, so we extended the cohort to seventeen on two preregistered criteria and no others, size-adjacency and publisher spread (three checkpoints were excluded on infrastructure grounds, never for their scores; Table 11 lists the cohort and Appendix F the rules). That gives **40** size-matched pairs, 32 of which cross publisher families; adapted BPB is right on 30 of those (0.938 lenient, 0.875 strict), so the result does not lean on any one publisher. Against the news criterion (Figure 4) adapted BPB agrees with the quiz on 0.900 of pairs under lenient scoring, ahead of every alternative, with HellaSwag next and the unadapted reading and parameter count behind it. Under strict scoring it is 0.850 and the unadapted reading is level with it; no alternative is ahead under either convention. Both generative benchmarks land *below* chance by point estimate, GSM8K at 0.359; their intervals still contain chance at this cohort size, so what is established is that neither is usable in this band, not that either is reliably worse than a coin. HellaSwag also clears chance at forty pairs, which the eleven-model cohort was too low-powered to show; the gap between it and adapted BPB is consistent but not yet resolved. Size, the natural baseline, is beaten outright: $+0.250$, 95% interval [$+0.074$, $+0.500$], positive in 99.4% of resampled cohorts under lenient scoring ($+0.100$ under

Criterion built from	scoring	Adapted BPB	Zero-shot BPB	HellaSwag	GSM8K	MMLU-Pro	Param. count
Google News (11)	lenient	0.909 [†]	0.545	0.818	0.182	0.182	0.636
	strict	0.727	0.727	0.636	0.364	0.364	0.818
Reddit (11)	lenient	1.000	0.636	0.909	0.273	0.273	0.727
	strict	0.909	0.727	0.818	0.364	0.364	0.818
Hacker News (10) (11)	lenient	0.900	0.800	0.800	0.500	0.600	1.000
	strict	0.727	0.818	0.636	0.545	0.727	0.818
arXiv math (10)	lenient	0.900	0.900	0.600	0.700	0.700	0.800
	strict	1.000	1.000	0.600	0.800	0.800	0.800

Table 3: The same comparison, run four times against four independent yardsticks. *Eleven-model cohort; 10–11 size-matched pairs per corpus* (count in parentheses; Hacker News has ten under lenient scoring and eleven under strict, because a pair the criterion ties orders nothing). Each row block scores every selector against the quiz built from *that* corpus’s own held-out text. Bold is the leader in the row; † marks an interval clearing chance. In every row adapted BPB is at or above every public benchmark, and no public benchmark tops any row; it holds the highest point estimate on two of the four corpora under both conventions—outright on Reddit, tied with the unadapted reading on arXiv—while ranking by size leads three rows. At ten or eleven pairs only one of the forty-eight cells clears chance, so the table is read for its orderings.

strict), and across all 136 pairs of the seventeen, not only the size-matched ones, $+0.088$ [$+0.016, +0.167$], lenient.

One quiz is one opinion, so Table 3 runs the whole comparison four times, each corpus judged only by the quiz built from its own held-out text: news, Reddit, Hacker News and arXiv mathematics. Two patterns hold on all four corpora under both conventions: adapted BPB never falls below a public benchmark in any row, and no public benchmark reaches the top of any row. Against every selector, size included, it leads or ties five of the eight. On news and Reddit both generative suites sit far below chance by point estimate—0.182 on the news criterion, picking the worse model four times out of five—while on Hacker News MMLU-Pro reaches 0.727 and on mathematics both recover to 0.700–0.800, in line with the aiming result of Section 3.2. Ranking by size scores higher in three rows—news strict and both Hacker News rows, at 0.818, 1.000 and 0.818—which at ten or eleven pairs is expected of a strong baseline.

4 Using the Measurement

In practice the measurement slots into a short procedure. When candidates differ by more than about $2\times$ in parameters, the larger one is usually the better choice; size is a good rule of thumb there, though adapted BPB beats it even across the full $70\times$ range under lenient scoring, and the protocol earns its cost where size has stopped deciding. Otherwise the text comes from the target domain, or close to it, dated after every candidate shipped and screened as in Phase 1; the smallest corpus we report on is 15 MB, and nearby text can serve too—on the ten pairs tested, Reddit BPB predicted the news fine-tune as well as news BPB did (Appendix E). Every finalist is then adapted by the same procedure and ranked by adapted BPB: the per-model search where compute allows, the hand-picked recipe where it does not, which at the costs of Phase 2 is cheap enough to run on every candidate rather than a shortlist; assembling the corpus, not the GPU time, is the main cost. The unadapted number is best read as a free screen, not the selector: it flags a candidate badly off-distribution for the target text, but alone reaches only 0.545 in the size-matched band (lenient; under strict the two readings are level).

With two or three candidates rather than a cohort, the *gap* is what matters, and we treat anything under

2% as a tie. A different seed moved adapted BPB by up to 1.24% (Gemma-4-12B; the median model, 0.01%) and token-aligned blocks by 0.18% (Appendix D). Of the 33 size-matched pairs at least 2% apart, the lower-BPB model was the criterion’s preference in **all** 33 lenient and 30 of 33 strict; closer than that, licence, latency or serving cost should decide. The threshold was chosen after seeing this cohort, so Table 9 is a calibration rather than an estimate. Results are comparable only when corpus, split, context, masking and adaptation all match.

5 Conclusion

Language models should be evaluated as language models: by the loss they were trained to minimise, in bits per byte, on fresh decontaminated text from the domain of use, after the same adaptation for every candidate. There is no ground truth to appeal to; what can be checked is whether the principled quantity agrees with the concrete uses we could build, and it agrees with each at least as closely as any benchmark, while each benchmark falls short on some. Adaptation changes the ordering; the new ordering agrees with HellaSwag at Spearman 0.96–0.98 on general text and turns to the reasoning benchmarks on mathematics; it matches the outcome of fine-tuning fifteen models more often than any benchmark, each within one rank of its finish; and on four corpora it never ranks candidates worse than a public benchmark. It needs only raw text and minutes of GPU time, runs on fresh screened text no benchmark can offer, and captures what a benchmark structurally cannot: how good a model becomes once adapted.

The natural objections were tested. *Why not simply fine-tune on the task?* When the task, its labels and the budget exist, that is the better evaluation for that task; a base model is usually chosen before any of them do, and must serve many tasks at once. *Circularity*—a yardstick built from the same corpus rewarding the method—is answered by the fine-tune, scored on separate documents against human-written references, whose ranking agrees with the criterion at Spearman 0.921. *Recipe dependence* is answered by repeating every fine-tune with a substantially different recipe, which leaves the ranking nearly unchanged (Spearman 0.982). *Contamination* is addressed by dating, deduplication and membership screens, and *selection bias* by extending the cohort on preregistered criteria and scoring every size-matched pair. What remains is power: at seventeen models the lead over HellaSwag is consistent but not yet resolved, and a larger cohort is the next step.

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A Related Work

From perplexity to comparable loss. Language modelling was grounded in byte- and character-level entropy from the start [1], but token-level perplexity depends on each model’s vocabulary [2, 4]; normalising by UTF-8 bytes restores a common unit [3, 5], and loss-based metrics carry better signal and lower noise than accuracy across 465 models [13]. Our contribution is the operational protocol around the metric, not the metric.

Where the loss is measured matters. Thrush et al. turn the varying correlation between per-domain perplexity and benchmark performance into a pretraining-data selection rule [14]. Their estimand is the complement of ours: they fix the benchmark suite and choose data to *train* one strong model, where we fix the candidate pool and ask how the *selection ranking* moves.

Equal preparation. Dynamic evaluation updates a model on recent test history [21, 22] and domain-adaptive pretraining improves fit to a target distribution [23]. Closest to our motivation, *train-before-test* gives 61 models identical supervised preparation across 24 tasks and raises cross-benchmark Kendall agreement from 0.52 to 0.76 [15]; they rank post-fine-tuning accuracy on public labelled benchmarks, where we rank held-out fit after *unsupervised* adaptation to a fresh corpus, across tokenizers. The core idea—prepare every candidate identically before comparing them—is theirs; what we add is doing it without labels, on text no vendor has seen, in a unit that compares tokenizers, and checking the result against real fine-tuned systems. Controlled families show a smooth loss-to-error relation [17, 16]; we test it *across* families.

Contamination. Models memorise duplicated pretraining text [6, 7], so deduplication and contamination analysis are necessary [8, 11]; reference-free membership signals are useful screens [9, 10] but prove nothing about a proprietary training set, so we use them only for conservative quarantine.

Corpus	Raw records	Documents	Median words	Text	Contamination control
Google News, 8 Jun 2026	119,054	119,054	362	361 MB	Min-K++ / context / perturbation union
Reddit, 6–10 Aug 2026 [†]	3,054,623	141,527	624	533 MB	Post-release scrape
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB	Post-release scrape
arXiv math. *, 9 Jun–20 Aug 2026 [‡]	3,207	3,207	—	15 MB	Post-release scrape + GSM8K shingle screen

Table 4: Corpora. All four receive identical exact-deduplication, Gopher quality filtering and MinHash/LSH near-duplicate removal at Jaccard 0.60; they differ in *contamination* control, deliberately. The reference-free screen ran on the news corpus, whose 8 June 2026 snapshot predates part of the cohort’s release window; Reddit and Hacker News were scraped after every model shipped, as was most of the arXiv set, and they are protected by construction. Reddit and Hacker News comments are grouped to document length. [†]Four days are available in the stated range. [‡]Used only for the mathematics adaptation.

B The Criterion: Construction and Composition

Table 5 gives the four criteria. Items come from each corpus’s own held-out test split, reproduced with the evaluation pipeline’s shuffle and seed. Each hides one entity or number span and supplies 1,200 characters of preceding context; spans occurring earlier in the prompt are dropped, so retrieval from context cannot substitute for prediction. Generation is greedy, scored by string match, and produced by the released base weights.

Composition differs across the four, which is why we never compare their levels: news is entity-dominated and Reddit number-dominated, and numeric spans are simply easier—answered at 0.279 and 0.325 against 0.180 and 0.060 for entity spans on those two corpora. Within a corpus every selector is scored on identical items, which is the only comparison the paper draws.

A worked item, answered by two of the cohort:

... MBDA is owned by Airbus (37.5%), BAE Systems (37.5%), and Leonardo (____
hidden span: 25% *Gemma-4-12B*: 25%) . ✓ *Qwen-2.5-0.5B*: 30%) . ×

The answer is nowhere in the context; the model has to know the company, or do the sum.

How precisely is the quiz known? We resample its 2,500 items, paired across models so that item difficulty cancels out of a between-model difference. It separates the cohort from 0.066 to 0.213 and resolves seven of the ten adjacent pairs under lenient scoring, two under strict; Table 1 carries every interval. Adjacent models it cannot order are reported as unordered rather than counted as some selector’s mistake.

Does sharing text with BPB make it circular? The criterion and the BPB evaluation do come from the same held-out bytes. The evidence that this does not drive the result is internal: the reading *mechanically closest* to the criterion—zero-shot BPB, same weights, same text, same conditional—is the weaker of the two at 0.545, while the adapted reading, from different weights, is the best. Shared bytes alone do not buy agreement, and Section 3.3 leaves the loop altogether: a real fine-tune, scored on different documents against a human-written reference.

Is it just measuring size? Such a criterion could also be a size proxy. Over the full $70\times$ range it does track size ($\rho = 0.918$ against log parameters), as any capability measure should. *Inside* the band, parameter count clears chance against none of the four per-corpus criteria, though it does on the extended news cohort under strict (0.750 [0.517, 0.932]), and two dense size inversions resolve—Gemma-4-12B over the larger Ministral-3-14B, +0.044 [+0.032, +0.057], and Llama-3.2-1B over Qwen-2.5-1.5B, +0.016 [+0.006, +0.026]. (The one mixture-of-experts model activates 3B of 34.7B and is never used as a size example.)

Screening. The news corpus additionally passes three reference-free membership screens per model—Min-K++ (normalised likelihood of the lowest-probability $K=20\%$ target tokens), context sensitivity (target NLL with and without sampled same-corpus prefixes), and, for flagged candidates only, preference over two word-swap/placeholder perturbations. The global union quarantined 80 unique sequences, after which every model receives the identical retained corpus. Near-100% preference for the unperturbed original is expected—fluency alone produces it—so we draw no memorisation conclusion from it (results/contamination/).

Criterion	Items	Entity	Number	Drawn from
Google News, 8 Jun 2026	2,500	1,677	823	test split of a 119,054-doc corpus
Reddit, 6–10 Aug 2026	2,500	900	1,600	test split of a 141,527-doc corpus
Hacker News, 6–10 Aug 2026	2,005	931	1,074	test split of a 6,086-doc corpus
arXiv math. *, 2026	1,911	949	962	test split of a 3,207-doc corpus

Table 5: The four criteria. Composition differs by corpus and that is why their *levels* are not comparable across rows: news is entity-dominated, Reddit number-dominated. Within a corpus the comparison is between selectors on identical items, which is the only comparison we make. The mathematics set is the thinnest, drawing on 320 held-out documents with up to eight spans from each.

Corpus	Reading	Rank correlation (ρ) with			GSM8K pairs
		GSM8K	MMLU-Pro	HellaSwag	correct
News	zero-shot	0.800	0.845	0.718	45/55
	adapted	0.700	0.773	0.982	42/55
Reddit	zero-shot	0.745	0.782	0.655	43/55
	adapted	0.655	0.745	0.964	41/55
Hacker News	zero-shot	0.800	0.809	0.627	44/55
	adapted	0.745	0.855	0.973	44/55
Math (arXiv)	zero-shot	0.764	0.736	0.482	43/55
	adapted	0.909	0.927	0.818	49/55

Table 6: *Eleven-model cohort, all 55 pairs (no size band).* All twenty-four (corpus $\times$ reading $\times$ benchmark) cells, recomputed from the committed cells. The rotation is visible column by column: adapting on general text raises HellaSwag agreement and lowers GSM8K everywhere, while adapting on mathematics reverses it.

C The Full Benchmark Alignment

Table 6 reports all twenty-four (corpus $\times$ reading $\times$ benchmark) cells over the full 55 pairs; Table 7 prices each reading’s top-1 pick against each benchmark’s own. The rotation (Appendix D) is legible column by column: adapting on general text raises HellaSwag agreement and lowers the GSM8K column on every corpus and MMLU-Pro on two of the three general-text corpora; adapting on mathematics reverses the GSM8K half. The pairs-correct column—GSM8K only—shows how little of this is visible in a rank correlation: across its eight rows it spans 41/55 to 49/55.

D Robustness and Measurement Artefacts

Stability across domains. All eleven models ran on three independently collected domains under the same recipe (Tables 4 and 12). Adaptation genuinely changes the choice—zero-shot and adapted rankings agree only at Spearman $\rho = 0.655$ on news, 0.609 on Reddit and 0.682 on Hacker News, with models moving up to seven places—and the adapted ordering then holds across domains: news and Reddit agree on nine of eleven exact ranks ($\rho = 0.991$, maximum displacement one), news against Hacker News gives 0.955 and Reddit against Hacker News 0.964, and the agreement is not merely ordinal (Figure 5). **So does the zero-shot ordering**—0.982, 0.964 and 0.973 on the same three comparisons. The tempting story is that adaptation buys stability, and it does not; it buys a *different* stable ordering, and Section 3.1 is the evidence that it is the better of the two.

Why the search gives every model the same number of trials. The search of Phase 2 uses a multivariate TPE sampler (seed 42) over learning rate (10^{-6} – 10^{-3} , log), physical batch (2, 4, 8), gradient accumulation (1–16), warm-up ratio (0–0.2), weight decay (10^{-5} – 10^{-1} , log), LoRA rank (8–64), α as a multiple of rank (1, 2, 4) and dropout (0–0.2); a successive-halving pruner starts trials at 50 steps and promotes the best third at each rung; and the objective is BPB

Corpus and reading	Picks	Accuracy given up vs. the benchmark’s own pick (pts)		
		GSM8K	MMLU-Pro	HellaSwag
News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
News adapted	Gemma-4-31B	1.6	4.3	0.0
Reddit zero-shot	Ministral-3-14B	5.8	8.6	5.0
Reddit adapted	Gemma-4-31B	1.6	4.3	0.0
Hacker News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Hacker News adapted	Gemma-4-31B	1.6	4.3	0.0
Math (arXiv) zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4

Table 7: What choosing on BPB costs you on each benchmark. For every corpus and reading, the model the ranking puts first and how many accuracy points that costs against each benchmark’s own top-1 model. Only three distinct picks occur in eight rows, which is itself the finding: the adapted ranking selects Gemma-4-31B on every general-text corpus—the HellaSwag leader outright, at 1.6 GSM8K and 4.3 MMLU-Pro points—while adapting on mathematics recovers the GSM8K and MMLU-Pro leader at zero regret. The corpus, not the metric, decides which column you pay in.

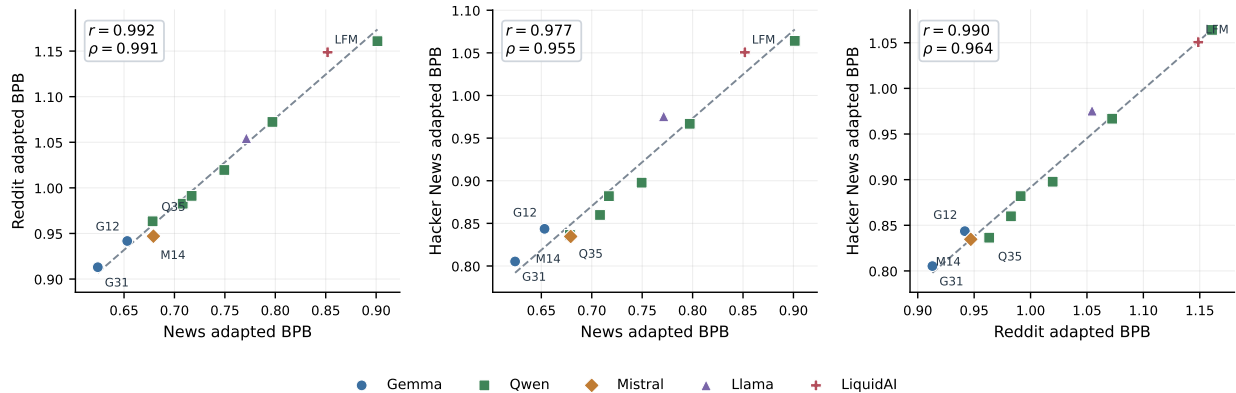


Figure 5: Pairwise adapted-BPB agreement across domains. Absolute difficulty changes, but points remain close to a monotone shared ordering: Pearson correlations between score values are 0.992 (news–Reddit), 0.977 (news–Hacker News) and 0.990 (Reddit–Hacker News). G31/G12 denote Gemma-4-31B/12B; Q35 is Qwen-3.5-35B-MoE; M14 is Ministral-3-14B.

on a fixed 256-block subset of the validation split. An earlier run of this search on the eleven-model news cohort was capped by wall-clock time rather than by trial count, and the cap was confounded with model size: the two largest models completed four trials each while the 1B models completed 17 and 18 (results/legacy_sweep/*/trials.csv), and the five models of 9B or more all kept the same third draw of the seeded sampler as their winner, while the smaller models’ winners came from much later in their searches. A time-capped search hands small models a better-tuned adapter and biases the ranking by parameter count, which is why the released pipeline budgets the search in trials and applies no time cap by default. None of those adapters produces a number in the paper; the reported results use the hand-picked configuration of Phase 2.

The cold start, seen a second way. Varying the inference context on 64 fixed held-out target windows from 16 to 2048 tokens reproduces Figure 2’s effect non-positionally: both models improve with context and the larger improves faster, so any protocol that scores short contexts under-rates the models that use long ones. Measured context sensitivity is 0.032 bits per byte per doubling, which is the constant the block-alignment bound below uses.

Token-aligned blocks: a real bias, bounded. Evaluation blocks are 512 *tokens* rather than a fixed byte span, so models with different tokenizers score slightly different amounts of text—5.92–6.24 MB across the eleven-model

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K
Gemma-4-31B	0.624 (1)	57.1 (2)	87.9 (1)	85.0 (3)
Gemma-4-12B	0.653 (2)	44.8 (7)	84.0 (3)	64.8 (7)
Qwen-3.5-35B-MoE	0.678 (3)	61.4 (1)	85.6 (2)	86.6 (1)
Ministral-3-14B	0.679 (4)	52.8 (4)	82.9 (4)	80.8 (6)
Qwen-3.5-9B	0.708 (5)	55.6 (3)	81.6 (5)	85.8 (2)
Qwen-2.5-7B	0.717 (6)	47.2 (6)	80.3 (6)	81.3 (4)
Qwen-3.5-4B	0.749 (7)	52.4 (5)	77.2 (7)	81.0 (5)
Llama-3.2-1B	0.771 (8)	11.2 (11)	65.8 (9)	6.4 (11)
Qwen-2.5-1.5B	0.797 (9)	25.6 (8)	68.0 (8)	61.6 (8)
LiquidAI-LFM2.5	0.852 (10)	16.4 (9)	60.9 (10)	54.1 (9)
Qwen-2.5-0.5B	0.901 (11)	16.0 (10)	51.8 (11)	34.3 (10)

Table 8: Where the readings disagree, in one place. *Eleven-model cohort, news.* Adapted news BPB (lower is better) beside the three benchmark scores, rank in parentheses. Gemma-4-12B is the case to look at: second by BPB, third on HellaSwag, and *seventh* on both MMLU-Pro and GSM8K—and second by the independent in-domain criterion (Table 1). Llama-3.2-1B and Qwen-2.5-1.5B are adjacent in BPB and 2.1 HellaSwag points apart, yet 55 GSM8K points apart.

cohort, a 5% spread, and 9.9% across the seventeen. Bounding the resulting bias with the measured context sensitivity puts it at $0.08\times$ the smallest adjacent rank gap in the eleven-model cohort. In the extended cohort the bound is looser and worth stating: on the worst adjacent pair, Qwen2.5-3B over Falcon3-7B, whose BPB gap is only 0.29%, the bound reaches $0.61\times$ the gap. It still **overturns no rank**, but byte-aligned blocking would remove the question, and we recommend it for future work.

The resolved comparison is not an artifact of a lucky seed. Every adaptation cell ran with the torch RNG unseeded, and the cluster bootstrap resamples models, so it carries no within-cell training variance. A preregistered check—three seed replicates for all 17 models, 51 cells, news corpus, everything else pinned—is confirmatory on all four questions it asked. Median within-model SD of adapted BPB is 0.000087 against a between-model SD of 0.074340, a ratio of 0.0012; in-band pairwise accuracy is 0.900 under each seed; *no* in-band pair—0 of 40—reverses its ordering; and adding a seed-resampling stage moves the resolved interval by 0.001. The spread is not uniform: Gemma-4-12B is the one model where seed noise is visible at all (SD 0.00315, $5.3\times$ the next-largest spread).

Steering: direction, not magnitude. Adapting on general text raises agreement with HellaSwag on news from $\rho = 0.718$ to 0.982 while GSM8K falls from 0.800 to 0.700 and MMLU-Pro from 0.845 to 0.773; adapting on contamination-screened arXiv mathematics instead raises GSM8K agreement to 0.909 and restores the GSM8K oracle pick at zero regret, while three general-text corpora tried as controls all leave GSM8K at or below the do-nothing baseline. That is the rotation, and it is why the body reports only its direction. Because the steering hypothesis predicts a *joint* movement, the better-powered statistic is a difference in differences—(GSM8K minus HellaSwag selection accuracy under mathematics adaptation) minus the same quantity under a general-text corpus—which is +0.242 pooled. Resampling models gives a strictly positive difference in 93.7% of draws; the stricter publisher-family clustering the mechanism itself implies gives 80.9% with not one negative draw in 20,000, but no longer clears 0.05 on any tie convention. A leave-one-family-out jackknife shows why: dropping the two Gemma models takes the pooled difference from +0.242 to +0.056. The remedy is more *families*, not more models.

Discarding the cold start: how much does it buy? (E10.) Each model was re-scored zero-shot with per-token loss accumulated separately at every position in the 512-token block, so every cutoff K is derivable from one pass; the falsifying outcome was fixed in advance—if any K brought the free reading within one in-band pair of adapted BPB, the mechanism claim above would have been withdrawn. It does not. Under lenient scoring in-band accuracy rises from 0.545 at $K=0$ to 0.636 at $K=128$ and no further, closing 25% of the 0.364 gap to adapted BPB. Under strict scoring the test is *uninformative* and we say so rather than counting it: the two readings already tie at 0.727 there, so there is no gap for the cold start to be accused of causing, and discarding positions only makes the free reading worse (0.636 at $K\geq 128$). The re-measured cells are not the published ones—they run up to 1.5% low—so every

cutoff is compared against this run’s *own* $K=0$, which reproduces the published zero-shot selector exactly under both conventions.

Scoring	Relative BPB gap	All in-band pairs		Criterion-resolved only	
		pairs	agreement	pairs	agreement
lenient	all	40	0.900	35	0.971
	$\geq 1\%$	38	0.947	35	0.971
	$\geq 2\%$	33	1.000	32	1.000
	$\geq 5\%$	22	1.000	22	1.000
strict	all	40	0.850	31	0.968
	$\geq 1\%$	38	0.895	31	0.968
	$\geq 2\%$	33	0.909	28	0.964
	$\geq 5\%$	22	0.955	19	1.000

Table 9: How often the lower adapted-BPB model is the better model, by how far apart they are. *Extended cohort: 17 models, 40 size-matched pairs.* Gaps are a percentage of the smaller adapted BPB. The measurement’s own floor is about 1.4% (seed variation plus block alignment); Section 4 rounds its rule up to the 2% row, where these counts begin.

Three reproduction controls. Re-running two Ministral-3-14B cells weeks later on different hardware reproduced zero-shot BPB exactly and adapted BPB to within 0.0003, two orders of magnitude below the differences compared. Because the extension models were adapted on a different software stack, a published cell (Qwen2.5-7B) was re-run through the extension pipeline; the analysis refuses to pool the cohorts unless it reproduces within 1%, and it reproduces to 0.01% on adapted BPB (0.7169 against 0.7170). And because benchmark rankings are themselves estimates, recomputing selection accuracy while discarding pairs whose benchmark gap is smaller than k combined standard errors leaves the result intact and slightly improves it: on GSM8K 0.909 at $k=0$ becomes 0.933 at $k=3$, and on HellaSwag 0.836 becomes 0.863, so the metric’s errors concentrate in pairs the benchmark itself cannot separate.

E The Downstream Fine-Tune

Section 3.3 reports the result; this is how it was built, and what it does and does not license. It ran in two stages, each preregistered in code before its job returned: first the two pairs on which the selectors disagree most (Gemma-4-12B against Ministral-3-14B, Llama-3.2-1B against Qwen-2.5-1.5B), then every cohort model that fits one GPU, so that no pair is chosen by anyone. Gemma-4-31B and Qwen-3.5-35B-MoE need more than one GPU *for this fine-tune*—its 1,280-token sequences and its generation pass cost far more memory than the 512-token adaptation priced in Section 1—and are excluded by that rule, which costs one in-band pair of forty.

The task. Journalistic convention makes a news article’s first paragraph a standalone summary of it, so the corpus supplies its own supervision: the input is the article with its opening paragraph removed, truncated to 3,000 characters, and the target is that paragraph. Articles qualify if they have at least three paragraphs, an opening paragraph of 200–700 characters ending in sentence punctuation, and at least 800 characters of remaining body. Of the 11,905 validation documents, 3,627 qualify; 2,000 are sampled for training and 500 for evaluation under a fixed seed.

Why the validation split. The in-domain criterion (Section 2.2) and the BPB test blocks both live in the *test* split. That is verified rather than assumed: the shipped criterion items reproduce item-for-item from an independent reconstruction of the test split, and the adaptation pipeline’s contamination pass—which would have split the corpus differently—is off by default and recorded as off in every published cell. Evaluation documents are then hashed against the test documents twice, on full normalised text and on a 400-character opening, the second because a syndicated wire story is republished with an edited body and an identical lede. Four articles were dropped by that second check and none by the first.

Equal preparation. Both members of a pair get the paper’s own adaptation recipe unchanged —LoRA rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, 250 steps—with prompt tokens masked out of the loss so the model is trained to write the lede rather than to reconstruct the article. Nothing is tuned on either model, because a per-model search is exactly the cost this protocol exists to avoid. The tokenizers differ, so the budget is matched in update steps $\times$ batch $\times$ sequence length; each model’s realised token count is in the artifact. Decoding is greedy with a 180-token cap and each checkpoint’s own generation configuration otherwise untouched, following Appendix G. The prompt is assembled so that the instruction always survives truncation: the *body* is cut to fit, never the instruction after it, because right-truncating the assembled prompt would delete the instruction from the longest articles at a different article count for each tokenizer—an asymmetry between the two models being compared. At most four of the 500 evaluation articles were cut for any model, and none for three of them.

The decision rules, fixed before each run. For the two chosen pairs: mean ROUGE-L F1 over the 500 held-out articles, a paired bootstrap over articles (20,000 draws, 95% percentile interval) and a practical-equivalence margin of 0.01—an interval excluding zero resolves the pair for whichever reading predicted that winner. The first pair resolved for adapted BPB; the second was inconclusive. For the full cohort: the fine-tuned ROUGE-L is treated as the criterion and every selector is scored with the paper’s own statistic—pairwise accuracy within $2\times$, model-level cluster bootstrap, 20,000 draws (Table 10). A separate guard voids any comparison in which fine-tuning failed to improve a model over its own base weights, since ordering unmoved models would order their base behaviour under another name. It did not fire: all fifteen models gained between $+0.041$ and $+0.153$.

A second recipe. Both readers of an earlier draft raised the same objection: the fine-tune reused the selector’s own recipe, so a model that adapts quickly under it could win both. Recipe B changes every setting the two shared—LoRA rank 64, $\alpha=128$, learning rate 3×10^{-4} , effective batch 16, 400 steps, a new seed—on the same task, data, decoding and metric, preregistered before it ran. It fits the training articles far harder and scores slightly below recipe A for every model, yet orders the fifteen systems almost identically (Spearman 0.982). Against recipe-B systems adapted BPB remains the top selector at 0.923 [0.750, 1.000], ahead of HellaSwag at 0.846, parameter count at 0.744 and GSM8K and MMLU-Pro near chance, and it picks the better system on all 20 pairs recipe B resolves. Its margin over parameter count, resolved under recipe A, does not resolve here ($+0.179$ [$-0.036, +0.441$]), so that margin should be read as recipe-specific (Table 10).

What copying buys. A news body restates its own lede, so reference overlap is partly available by retrieval. Two reference points bound it: the article’s first body sentence scores 0.125 against the reference and a random body sentence 0.111. Unadapted base weights land close to copying—inspection confirms they largely do copy—while every fine-tuned system clears it, from 0.188 (Qwen-2.5-0.5B) to 0.249 (Gemma-4-12B).

	All 39 pairs	95% CI	22 resolved pairs	Recipe B
Adapted BPB	0.949	[0.817, 1.000]	0.955	0.923
HellaSwag	0.872	[0.640, 1.000]	0.909	0.846
Zero-shot BPB	0.769	[0.444, 1.000]	0.682	0.744
Parameter count	0.718	[0.463, 0.912]	0.773	0.744
MMLU-Pro	0.513	[0.250, 0.846]	0.500	0.538
GSM8K	0.474	[0.194, 0.824]	0.455	0.447

Table 10: Which selector predicts the fine-tuned system? Fifteen models, each fine-tuned identically on the lede task; the criterion is each system’s ROUGE-L on 500 held-out articles. Each row is a selector’s pairwise accuracy over every size-matched pair, with the model-level cluster bootstrap, and its accuracy on the 22 pairs whose fine-tuned difference is itself resolved by a paired bootstrap over articles. The last column scores the same selectors against systems built with the deliberately different recipe B. GSM8K ties one pair and is scored on 38.

How close the corpus has to be. Exploratory, run after the result at a reader’s request: scoring the news fine-tune with adapted BPB from each of the other corpora, on the ten in-band pairs of the nine fine-tuned models that have all four, gives 0.900 for news and for Reddit, 0.700 for Hacker News and 0.600 for arXiv mathematics; the unadapted

readings reach 0.500–0.600. Fresh general text does most of the work and nearby text does it best. Ten pairs support no interval, so this is a direction, not a result.

What this does not show. One task, one domain, two recipes, one seed each. On this task adapted BPB ordered the fine-tuned systems better than any other selector and GSM8K and MMLU-Pro ordered them at chance; its margin over HellaSwag does not resolve. It is not a claim that adapted BPB predicts downstream utility in general, and the paper does not make one.

F The Cohorts

Table 11 is the extended 17-model cohort behind Figure 4 and Section 3.4: every model, its parameter count, its adapted news BPB and its score on the in-domain criterion with that criterion’s own interval. Table 12 then gives the original eleven across all three general corpora and both BPB endpoints, which is the population every other analysis in the paper uses. Table 12 is the full three-corpus leaderboard: both endpoints for all eleven models on news, Reddit

Model	Params (B)	Adapted BPB	In-domain criterion		Cohort
			score	95% CI	
Gemma-4-31B	31.4	0.624	0.213	[0.197, 0.229]	original
Mistral-Nemo-12B	12.3	0.646	0.166	[0.152, 0.181]	added
Gemma-4-12B	12.0	0.653	0.186	[0.171, 0.201]	original
Qwen-2.5-14B	14.8	0.675	0.137	[0.124, 0.151]	added
Qwen-3.5-35B-MoE*	34.7	0.678	0.147	[0.134, 0.161]	original
Ministral-3-14B	14.0	0.679	0.142	[0.128, 0.155]	original
Qwen-3.5-9B	9.0	0.708	0.116	[0.104, 0.129]	original
Qwen-2.5-7B	7.7	0.717	0.121	[0.108, 0.134]	original
Falcon3-10B	10.4	0.728	0.094	[0.083, 0.106]	added
Qwen-3.5-4B	4.2	0.749	0.101	[0.089, 0.112]	original
Qwen-2.5-3B	3.1	0.756	0.107	[0.095, 0.119]	added
Falcon3-7B	7.5	0.758	0.086	[0.075, 0.097]	added
Llama-3.2-1B	1.2	0.771	0.104	[0.092, 0.116]	original
SmolLM2-1.7B	1.7	0.788	0.091	[0.080, 0.102]	added
Qwen-2.5-1.5B	1.6	0.797	0.089	[0.078, 0.100]	original
LiquidAI-LFM2.5	1.2	0.852	0.078	[0.067, 0.088]	original
Qwen-2.5-0.5B	0.5	0.901	0.066	[0.057, 0.076]	original

Table 11: The extended cohort, in full. All 17 models ordered by adapted news BPB, with the criterion score each is judged against and its paired item-bootstrap interval. “added” marks the six models admitted for the size-matched extension (Section 3.4); the other eleven carry the mechanism, four-corpus and stability analyses. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters; parameter count here is always the total.

and Hacker News, ordered by adapted news BPB. It is the artifact every eleven-model number in the paper derives from, as Table 11 is for the seventeen.

The study ran at eleven models before the extension, which supersedes those numbers without contradicting them: selection accuracy exceeded chance in 23 of the 24 (corpus, reading, benchmark) cells—strongest for adapted BPB against HellaSwag on Hacker News (0.964, CI [0.840, 1.000]) and on news (0.964, CI [0.875, 1.000]), with mathematics-adapted BPB against GSM8K at 0.891 [0.725, 1.000], the one exception being mathematics-corpus *zero-shot* BPB against HellaSwag (0.727, CI [0.478, 0.980]). Between-corpus contrasts did not separate (+0.127 on GSM8K and MMLU-Pro, −0.127 on HellaSwag, every interval containing zero), and adapted BPB was the only in-band selector clearing the in-domain criterion (0.909, against 0.818 HellaSwag, 0.636 parameter count and 0.545 unadapted). Eleven models give eleven in-band pairs, and at that size a selector must be near-perfect to separate—which is what the extension in Section 3.4 fixed, and why the earlier claim that no benchmark clears chance was a statement about power rather than about benchmarks. Three checkpoints are excluded. The two OLMo-2 models fell

Model	News		Reddit		Hacker News	
	zero-shot	adapted	zero-shot	adapted	zero-shot	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LiquidAI-LFM2.5	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 12: Marker-masked BPB under one fixed adaptation protocol. Lower is better. Rows are ordered by adapted news BPB; all eleven models occur in all three corpora.

under the preregistered infrastructure-fault rule: their adapted BPB cells exist and are unused, because the criterion’s generation engine would not initialise for them, identically before and after a cache rebuild. The third, Qwen-2.5-32B, did not finish its 250 steps inside the run window; it would have added about two in-band pairs at the top of the size range. The OLMo exclusion removed a whole publisher, which is what the cohort most needs.

G The Benchmark Column Re-Score

Harness configuration. MMLU-Pro is 5-shot chain-of-thought with exact match on one stratified 1,000-question subset (seed 42); HellaSwag is 10-shot length-normalised multiple-choice accuracy on all 10,042 validation examples; GSM8K is 5-shot chain-of-thought with flexible numeric extraction on all 1,319 test examples. All bf16, greedy, tokenizer-native BOS.

July’s eleven-model column was scored on a host that no longer exists and generative scores do not pool across hosts—identical released weights move -0.60 to $+0.69$ GSM8K points when only the host changes, A100 to H100—so it was re-scored in full—seventeen models $\times$ three benchmarks = fifty-one cells, one host (A100-SXM-64GB), one build—with the design and the exclusion rule fixed in advance. Task definitions match exactly (17/17 identical task hashes on both hashed benchmarks) and every overlapping cell agrees with July to within 1.3 points, most to within 0.3.

Reaching that took one correction worth reporting, because the fault was in our own fix. Because the newer backend honours each checkpoint’s shipped `generation_config.json` and the original did not, we replaced every checkpoint’s generation settings with one identical policy—which also discarded `suppress_tokens`, a setting that is live for gemma-4-12B. That single omission cost it 9.86 GSM8K and 14.30 MMLU-Pro points while leaving HellaSwag, scored by likelihood and never calling `generate()`, intact. Keeping the setting for all seventeen models reproduces July to $+1.21$ points on GSM8K, $+0.10$ on MMLU-Pro and -0.27 on HellaSwag. That GSM8K cell is the one outside the preregistered 1.0-point tolerance, within the allowance of one that the re-score declared in advance: the column is within its stated allowance rather than clean, and those are different claims. The lesson generalises: a wrapper imposed to remove a confound can be the confound, so the unmodified harness belongs in the design as a control.